E-Commerce Performance Analysis

Identifying sales performance, product contribution & refund behaviour

1. Introduction

What This Dataset Is About

This dataset represents an e-commerce business and includes data related to website sessions, customer orders, order items, products, and refunds. It is used to analyse overall business performance by examining sales, revenue contribution, pricing patterns, customer purchasing behaviour, and refund trends across different products.

What Problem We Are Solving

The problem being addressed is understanding and improving e-commerce performance by identifying key revenue drivers, evaluating product-level performance, and detecting areas of revenue leakage such as high refund rates.

The goal is to generate actionable insights that help optimize sales, reduce refunds, and improve overall business decision-making.

2. Objectives / Analysis Questions

The following core questions drive this analysis:

- What is the overall sales performance in terms of total revenue, average order value, and conversion rate?
- Which products contribute the most to revenue and sales volume?
- What are the refund patterns across different products, and which products have the highest refund rates?
- Is there any relationship between product pricing and refund behaviour?
- How concentrated is the revenue across products (dependency on top-performing products)?

3. Dataset Information

Source

Maven Analytics – csv form

Key Tables

- Orders
- Products
- Order items
- Refunds
- Website sessions

4. Tools used

Excel – Raw dataset handling
SQL – Data extraction, joins, aggregation and KPI calculation
Python – Data cleaning, validation and preprocessing
Power BI – Dashboard creation & visualisation

5. Key KPIs

- Total Revenue – $1.94M
- Average Order Value (AOV) – $59.99
- Refund Rate – 14.66%

6. Dashboard Overview

The dashboard provides a structured view of e-commerce performance:

- **Top KPIs** – Quick summary of revenue, average order value, and refund rate.
- **Revenue Analysis** – Product-wise contribution to total revenue.
- **Sales Volume Analysis** – Comparison of units sold across products.
- **Refund Analysis** – Identification of products with high refund rates.
- **Pricing Insights** – Average price comparison across products.
- **Refund Distribution** – Share of total refunds contributed by each product.
- **Product Filter** – Interactive selection to analyse individual product performance.

7. Key Insights

- The Original Mr. Fuzzy dominates performance, contributing the highest revenue (~$1.21M) and sales volume (~24K units).
- Revenue is highly concentrated, with one product driving a major share, indicating dependency risk.
- The Birthday Sugar Panda has the highest refund rate (~6.04%), highlighting potential issues with product quality or customer expectations.
- Higher-priced products show relatively lower refund rates, suggesting better perceived value or satisfaction.
- Lower-priced products tend to have higher refund rates, contributing to revenue leakage.
- A significant portion of total refunds (~72%) comes from The Original Mr. Fuzzy due to its high sales volume.
- Moderate-performing products like The Forever Love Bear and The Hudson River Mini Bear contribute less to revenue but maintain comparatively lower refund rates.

8. Business Recommendations

- Focus marketing and inventory efforts on The Original Mr. Fuzzy to maximize revenue, while reducing over-dependency by promoting other products.
- Investigate the high refund rate of The Birthday Sugar Panda to identify issues related to quality, packaging, or product description.
- Improve product descriptions, images, and customer expectations to reduce refund rates, especially for lower-priced products.
- Optimize pricing strategies by balancing affordability with perceived quality to minimize refunds and maximize profit.
- Diversify the product portfolio to reduce reliance on a single high-performing product and improve overall stability.
- Continuously monitor refund rates as a key indicator of customer satisfaction and product performance.

9. Conclusion

The analysis highlights that while the business generates strong revenue, it is heavily dependent on a single top-performing product. At the same time, refund patterns reveal potential issues in certain products, leading to revenue leakage. By addressing high refund rates, optimizing pricing and product quality, and diversifying the product portfolio, the business can improve overall performance, reduce risks, and achieve more sustainable growth.